DATASET INTELLIGENCE & OPERATIONAL ANALYSIS

Domain: Universal Dataset Intelligence & Distribution Analysis (0.0%) | Profiling Conf: 89.6% | Records: 350 | Hash: 38d2e9b4

Filter: ALL_RECORDS(UNFILTERED) | Audit: VERIFIED_DETERMINISTIC_PARITY | Engine: v8-javascript

KEY PERFORMANCE INDICATORS (DETERMINISTIC METRIC AUDIT)

TOTAL RECORDS VOLUME

350

`COUNTROWS`

Omitted: 0 rows

UNIQUE ENTITIES (STUDENT_ID)

350

`COUNTDISTINCT(Student_ID)`

Omitted: 0 rows

TOTAL CURRENT TUITION_BALANCE_US

119,712

`SUM(Tuition_Balance_USD)`

Omitted: 0 rows

MEAN TUITION_BALANCE_USD

342.03

`AVERAGE(Tuition_Balance_USD)`

Omitted: 0 rows

MEDIAN TUITION_BALANCE_USD

0

`MEDIAN(Tuition_Balance_USD)`

Omitted: 0 rows

MEAN CUMULATIVE_GPA

3.06

`AVERAGE(Cumulative_GPA)`

Omitted: 0 rows

MEDIAN CUMULATIVE_GPA

3.10

`MEDIAN(Cumulative_GPA)`

Omitted: 0 rows

EXECUTIVE MANAGEMENT DECISION BRIEF

- **Ingestion & Data Quality Profile:** Audited 350 source rows across 7 columns with 100.0% full record completeness. Detected 0 duplicate records in source data.
- **Total Records Volume Benchmark:** Total Records Volume stands at 350 calculated via COUNTROWS. Supporting benchmark Mean Tuition_Balance_USD stands at 342.03.
- **Cohort Concentration (Academic_Major):** Largest cohort resides in 'Biochemistry' with 24,804 (20.7% of cohort total). Lead cohort exceeds 'Economics' by 1.1x.
- **Unique Entities (Student_ID) Distribution:** Distribution of Unique Entities (Student_ID) of 350. Denominator rules and null exclusions applied deterministically.

DATASET INTEGRITY & STATISTICAL COVERAGE

Audited Records: 350 | Fully Populated: 350 (100.0%) | Duplicates: 0 | Missingness Exclusions Documented

VISUAL COHORT BREAKDOWNS & OPERATIONAL ACTION QUEUE

DISTRIBUTION BY ACADEMIC_MAJOR (Academic_Major vs Total Tuition_Balance_USD)

VOLUME BY CLASS_STANDING (Class_Standing vs Volume)

OPERATIONAL ACTION QUEUE (DETERMINISTIC SOURCE TRACEABILITY)

RANK	ENTITY ID	SRC ROW #	PRIORITY	STATUS	ASSIGNEE / SITE	TRIGGER REASON
#1	STU-000004	Row #4	High	Active	— / —	Outstanding Balance: 1,204
#2	STU-000008	Row #8	High	Active	— / —	Outstanding Balance: 1,208
#3	STU-000012	Row #12	High	Active	— / —	Outstanding Balance: 1,212
#4	STU-000016	Row #16	High	Active	— / —	Outstanding Balance: 1,216
#5	STU-000020	Row #20	High	Active	— / —	Outstanding Balance: 1,220
#6	STU-000024	Row #24	High	Active	— / —	Outstanding Balance: 1,224
#7	STU-000028	Row #28	High	Active	— / —	Outstanding Balance: 1,228
#8	STU-000032	Row #32	High	Active	— / —	Outstanding Balance: 1,232
#9	STU-000036	Row #36	High	Active	— / —	Outstanding Balance: 1,236
#10	STU-000040	Row #40	High	Active	— / —	Outstanding Balance: 1,240

METHODOLOGY, AUDIT TRAIL & METRIC DICTIONARY

METHODOLOGICAL POLICIES & AGGREGATION AUDIT TRAIL

Policy #1: All analytical aggregations are computed locally in-browser using full-precision raw IEEE 754 float values.

Policy #2: No intermediate rounded values are ever re-aggregated across groups, metrics, or export layers.

Policy #3: Non-additive metrics (ratings, GPA, percentages, unit prices) are aggregated via weighted or arithmetic mean/median only; summation is strictly forbidden.

Policy #4: Semi-additive metrics (stock on hand, account balances) are point-in-time state indicators and must not be aggregated over time dimensions.

Policy #5: Missing, null, or empty string values are excluded on a per-metric basis without corrupting adjacent columns.

Policy #6: Zero AI hallucination guarantee: 100% deterministic code compilation with verified three-way engine parity.

CANONICAL METRIC DICTIONARY & ADDITIVITY TIERS

METRIC LABEL	FORMULA SPECIFICATION	ADDITIVITY	EXCLUSIONS	CANONICAL VALUE
Total Records Volume	`COUNTROWS`	**ADDITIVE**	0 omitted	**350**
Unique Entities (Student_ID)	`COUNTDISTINCT(Student_ID)`	**ADDITIVE**	0 omitted	**350**
Total Current Tuition_Balance_	`SUM(Tuition_Balance_USD)`	**SEMI_ADDITIVE**	0 omitted	**119,712**
Mean Tuition_Balance_USD	`AVERAGE(Tuition_Balance_USD)`	**NON_ADDITIVE**	0 omitted	**342.03**
Median Tuition_Balance_USD	`MEDIAN(Tuition_Balance_USD)`	**SEMI_ADDITIVE**	0 omitted	**0**
Mean Cumulative_GPA	`AVERAGE(Cumulative_GPA)`	**NON_ADDITIVE**	0 omitted	**3.06**
Median Cumulative_GPA	`MEDIAN(Cumulative_GPA)`	**NON_ADDITIVE**	0 omitted	**3.10**

INDEPENDENT MATHEMATICAL RECONCILIATION & REPRODUCIBILITY GUARANTEE

Source Hash: 38d2e9b4 | Report Hash: 9157b073 | Audit Timestamp: 2026-10-23T16:31:01.406Z

All metrics verified against independent Python float64 recalculation. Full raw precision preserved across all export targets.